



USA TRIATHLON EDUCATION

Regression Rates in Running

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Overview

Regression rate is the percentage your running speed drops as we move from one set distance to another distance that is double the first. It shows how well an athlete holds speed as the effort gets longer, and it points straight at what to train next.

The average regression rate for endurance athletes is about 6.5%. A rate well above that suggests the aerobic base is the limiter. A rate well below it suggests the athlete is aerobically strong but short on speed and speed endurance.

You get it from two field tests over distances where the second is double the first, which makes it easy to repeat as often as you retest.

How to calculate it

Run a maximal effort over a set distance, then a maximal effort over double that distance (make sure athletes have been signed off by their doctor for physical exercise, and only complete these tests once the athlete is well conditioned to training). Double the first time so it is comparable to the second, then measure how much slower the longer effort was.

My personal choice is normally 800 m and 1600 m, because both can be completed in one session and the results are relatively accurate for both short and long course athletes. The variances, and the reasons for them, are in my notes on running biomechanics at pace.

Worked example from the running calculator, using 800 m and 1600 m:

Step	Value
800 m test	2:38
Converted 800 m (doubled)	5:16 (316 s)
1600 m test	5:37 (337 s)
Regression rate	6.65% (0.0665 as a decimal)

That athlete sits almost exactly on the 6.5% average regression rate for endurance athletes. The Running Pace and Training Zone Calculator does this conversion automatically and shows the result as both a percentage and a decimal.

What the number tells you

Regression rate	What it suggests	Training focus
8 to 9% or higher	Insufficient aerobic conditioning	Build aerobic base and endurance
About 6.5%	Typical, balanced profile	Keep development balanced
3 to 4% or lower	Good aerobic base, limited speed	More speed and speed endurance work

Regression rate works best as part of a set of tests. The 400 m test shows speed and anaerobic capacity, the 800 m shows speed endurance and VO2 max capability, and the 1600 m shows aerobic power and threshold performance. Comparing performance across those distances shows whether an athlete's development is balanced or lopsided.

Setting paces and predicting races

The 4 x 1600 m test pairs with regression rate. Run four 1600 m repeats at roughly 90 to 100% effort (make sure athletes have been signed off by their doctor for physical exercise, and only complete this test once the athlete is well conditioned to training) with 90 seconds of recovery. The average pace across the four predicts 5K race pace.

That prediction has proven accurate: 5K to 10K times within a couple of seconds, and half marathon finish times within a couple of minutes. It suits short-course and 70.3 athletes especially well.

In the calculator, the predicted 5K time becomes the baseline and the athlete's own regression rate scales it to other race distances, including a custom distance. The same inputs drive the speed-endurance goals, such as 2 km targets built from the 5K prediction.

Use the regression rate to decide where the next block goes. A high rate sends the athlete toward aerobic conditioning. A low rate sends them toward speed and speed endurance.

Limitations and coaching notes

- **Treat 6.5% as a reference, not a target.** It is the average regression rate for endurance athletes. What matters is where the athlete sits and which way it moves between tests.
- **Test fresh.** Run the tests in the first week after an adaptation week, when the athlete is rested and adapted, and keep conditions consistent from test to test.
- **Both efforts must be true maximal efforts.** The rate only means something if both efforts were given everything (make sure athletes have been signed off by their doctor for physical exercise, and only complete these tests once the athlete is well conditioned to training).
- **Pair it with mechanics.** Technique breakdown tends to show up at higher speeds, and form changes such as lowering vertical oscillation reduce fatigue and help an athlete hold pace later in a race. Video analysis during progressively faster intervals is worth adding.
- **Use it alongside other data.** Combine it with the 400 m and 4 x 1600 m tests and with LT1, LT2, and VO2 information for a full picture of the athlete.